

ATOMIC MODELS

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Chapter 1

ATOMIC MODELS

SENIOR 4: TERM 1

Theme: Modern Physics

TOPIC 5: ATOMIC MODELS

16 PERIODS

Competency: The learner should be able to appreciate the different atomic models and how they are used to explain the basic structure of atoms.

LEARNING OUTCOMES	SUGGESTED LEARNING ACTIVITIES	SAMPLE ASSESSMENT STRATEGY
The learner should be able to:		
a. understand the structure of an atom in terms of a positive nucleus and negative electrons (u) b. understand the terms: atomic number, mass number, and isotopes, and use them to represent different nuclides (k, u) c. understand the methods by which electrons are ejected from /matter atoms and how these electrons are useful (u, v/a)	• In pairs, learners revisit prior learning about atoms, (atomic number, mass number, nuclides, isotopes) and research on the representation of different atoms • In groups, learners research and compare the Dalton model and Rutherford model of the atom. Explain atomic structure and components of a range of elements and isotopes in a report and on a poster. • In groups, learners research on and explain in a presentation: • how thermionic emission and the photoelectric effect occur • how cathode rays and X-rays differ	• Listen to group and pair discussions and pose questions to check and secure knowledge and understanding. • Evaluate quality of learning and progress towards the learning outcomes through assessment of reports and presentations.
Note: 1. Laws of photoelectric effect are not required 2. Numerical problems related to acceleration of electrons are beyond this level.		
ICT Support Learner watches computer simulations about thermionic emission and photoelectric effect.		

STRUCTURE OF AN ATOM

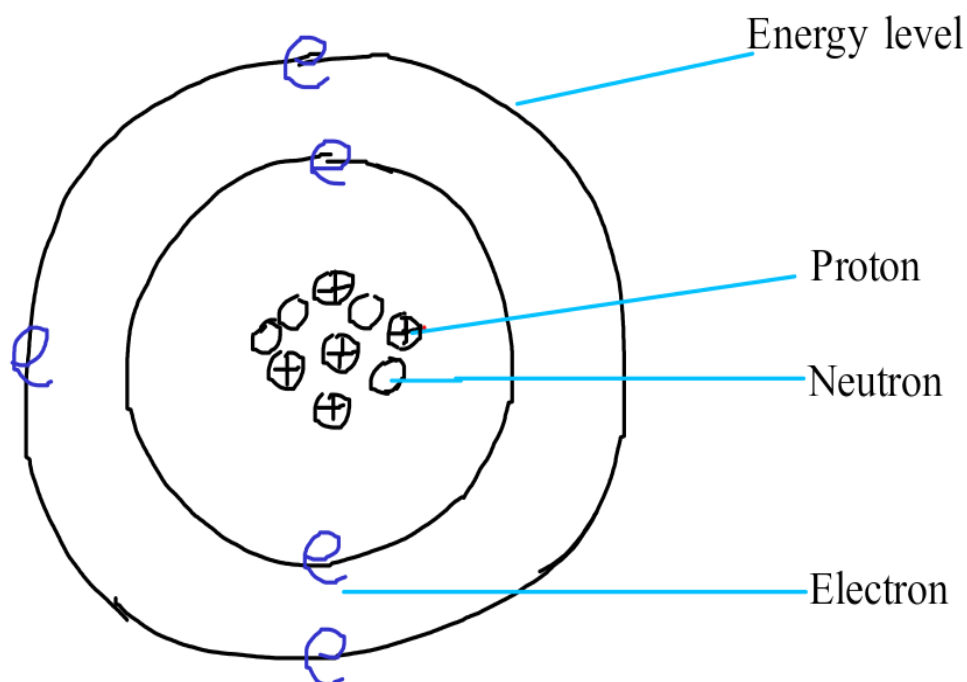
An atom is the smallest unit of an matter.

Atoms are the building blocks of all matter. Atoms can be combined with other

atoms to form molecules.

Definition: An atom is the smallest electrical neutral particle of an element that can take part in a chemical reaction.

QN: Draw a well labelled structure of an atom and describe it.



An atom is made up of three sub atomic particles namely:

- Electrons
- Protons
- Neutrons

The nucleus is the central part of an atom and consists of protons and neutrons.

Protons carry a positive charge while neutrons carry no charge.

Electrons are negatively charged and revolve in orbits around the nucleus.

Note:

- (i) Protons and neutrons are found in the nucleus and they are called **nucleons**.
- (ii) The nucleus of an atom is positively charged because it has protons that are positively charged and neutrons that carry no charge.
- (iii) The atom is neutral since the number of protons is equal to the number of electrons.

Characteristics of sub-atomic particles

Protons

- They are positively charged.
- They are heavier than electrons.

Neutrons

- They have no charge.
- The mass of neutrons is almost the same as that of protons.

Electrons

- They are negatively charged.
- They have negligible mass.

DEFINITIONS

Atomic number

This is the number of protons in the nucleus of an atom. It is denoted by letter Z.

Mass number/ Atomic number/ Nuclear number

It is the number of protons and neutrons in the nucleus of an atom. It is denoted by letter A.

If n is the number of neutrons in the nucleus, then:

$$\text{Mass number} = \text{Atomic number} + \text{Number of neutrons}$$

$$A = Z + n$$

Nuclide:

This is a distinct kind of an atom characterized by a specific number of protons and neutrons.

Nucleon:

This is the collective term to protons and neutrons in the nucleus of an atom.

ISOTOPES

Isotopes are atoms of the same element with the same number of protons and electrons but different number of neutrons.

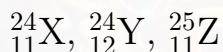
An atom is symbolically represented by A_ZX where X is the element, A is the mass number and Z is the atomic number.

Isotopes of an element have different mass numbers

Examples of isotopes include carbon-14

QNS:

1.(a) Which of the following nuclei belong to the same element.



(b) State and explain the nuclei given to such nuclei.

X and Z are isotopes. This is because they have the same number of protons.

(c) What is the atomic number of element Z.

(d) What is the mass number electrons of element Y.

(e) What is the number of electrons of element Y.

(f) What is the number of neutrons of elements Y and Z.

3. An atom of cobalt has an atomic number of 27 and a mass number of 59. How many protons and neutrons does it have?



Protons = 27

Neutrons = $59 - 27 = 32$

4. Uranium-235 and Uranium-238 are isotopes of Uranium and have the same atomic number 92.

(a) What do the numbers 235 and 238 represent?

Mass number

(b) What does the number 92 tell you about the nucleus of either of these atoms?

The nucleus has 92 protons

(c) What else does the number 92 tell you about the atom as a whole?

The atom has 92 electrons.

(d) In what way does the nucleus of Uranium-235 differ from the nucleus of Uranium-238?

Uranium-235

Number of neutrons = $235 - 92 = 143$

Uranium-238

Number of neutrons = $238 - 92 = 146$

∴ Uranium-238 has more neutrons than Uranium-235

5. Identify some elements that exist with their isotopes.

Element	Structure
Chlorine-35, Chlorine-37	$^{35}_{17}\text{Cl}$, $^{37}_{17}\text{Cl}$
Hydrogen-1, Hydrogen-2, Hydrogen-3	^1_1H , ^2_1H , ^3_1H
Carbon	$^{12}_6\text{C}$, $^{13}_6\text{C}$, $^{14}_6\text{C}$
Oxygen	$^{16}_8\text{O}$, $^{17}_8\text{O}$, $^{18}_8\text{O}$
Helium	^3_2He , ^4_2He
Uranium	$^{235}_{92}\text{U}$, $^{238}_{92}\text{U}$
Nitrogen	$^{14}_7\text{N}$, $^{15}_7\text{N}$
Neon	$^{20}_{10}\text{Ne}$, $^{21}_{10}\text{Ne}$, $^{22}_{10}\text{Ne}$
Magnesium	$^{24}_{12}\text{Mg}$, $^{25}_{12}\text{Mg}$
Silicon	$^{28}_{14}\text{Si}$, $^{29}_{14}\text{Si}$, $^{30}_{14}\text{Si}$

Some isotopes give radiation through a nuclear change and they are known as **radioactive isotopes**. While others do not give off any radiations and they are known as stable isotopes.

All isotopes with an atomic number greater than 82 are radioactive.

NB: Exposure to radioactive isotopes is harmful except in small quantities.

Exercise

1. Uranium U has 92 electrons and 146 neutrons. Write down the Symbol of Uranium nuclide.



2. Carbon C has 6 protons and 6 neutrons in its nucleus. Write down the symbol of carbon nuclide.



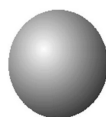
3. Why do Isotopes go through similar chemical reactions?

Isotopes are of the same element and have the same number of protons and electrons.

ATOMIC MODELS

An atomic model is a scientific representation that explains the structure of an atom. It shows how particles in an atom are arranged and how they behave.

1. DALTON'S ATOMIC MODEL



Dalton's atomic model was the first complete attempt to describe all matter in terms of atoms and all their properties.

Postulates of Dalton's Atomic model

1. All matter is composed of indivisible particles called atoms.
2. Atoms cannot be created or destroyed.
3. Atoms of the same element are similar in size, mass and shape but different from atoms of other elements.
4. Atoms of different elements may combine with elements form compounds.
5. The atom is the smallest unit of matter that can take part in a chemical reaction.

Successes of Dalton's atomic model

1. It provides a basis to distinguish between elements and compounds.
2. Dalton's theory explains the laws of chemical combinations eg law of conservation of mass, law of constant proportions , etc.

Failures of Dalton's Atomic model

1. The indivisibility of an atom was proved wrong.
2. An atom can further be sub-divided into protons, electrons and neutrons.
3. His model was unable to explain the existence of isotopes.
4. He did not explain chemical bonding.
5. He did not account for existence of allotropes.

2. RUTHERFORD ATOMIC MODELS

Rutherford's model defined an atom as a small, tiny, dense and positively charged nucleus at the center. This model is more widely accepted compared to Dalton's model.

Postulates of Rutherford's and atomic model

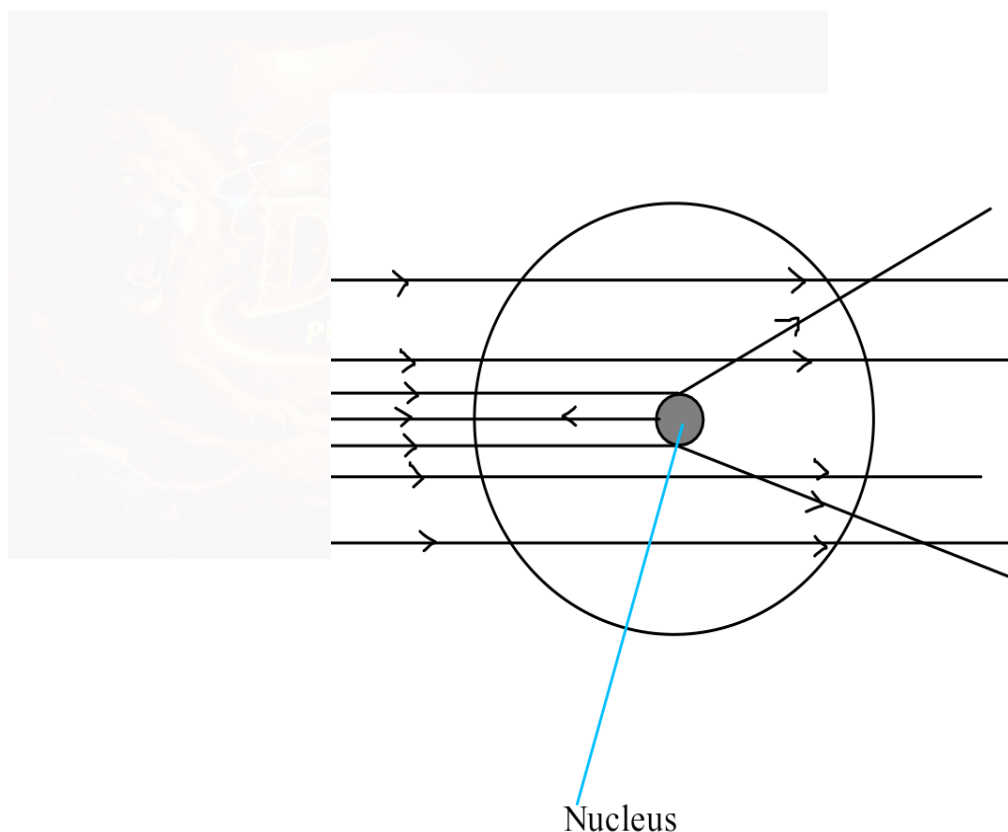
1. The atom has a small dense and positively charged nucleus at its

center.

2. The nucleus contains most of the atomic mass.
3. The electrons orbit the nucleus at a distance and are much lighter than the nucleus.
4. Electrons are negatively charged and occupy a large volume of space compared to the nucleus.
5. The positive charge of the nucleus is balanced by the negative charge of the electrons.
6. The atom is mostly empty space with electrons and a nucleus being very small compared to the overall size of the atom.

Successes of Rutherford's atomic model.

1. He explained the large angle scattering of alpha particles by gold foil which was previously unexplained.



Alpha particles that were far from the nucleus were undeflected.

Alpha particles that passed close to the nucleus were deflected at very large angles.

Alpha particles that were directed towards the nucleus were bounced off along the same path.

2. He provided a clear understanding of the structure of atoms with a small, dense nucleus and orbiting electrons.
3. He predicted the existence of neutrons which were later discovered.
4. He explained the properties of elements such as atomic mass and atomic number.
5. His model led to the development of nuclear physics and nuclear energy.

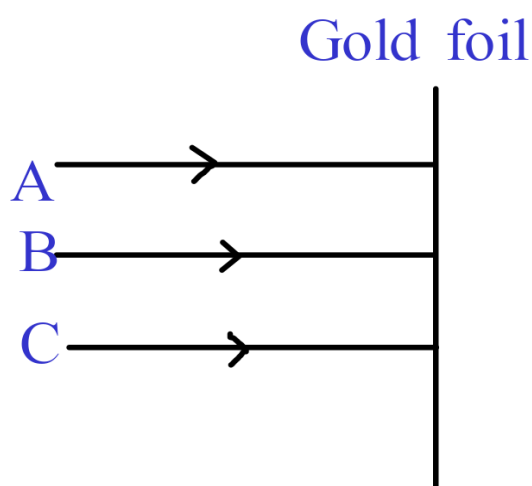
Conclusions from Rutherford's atomic model.

1. The biggest part of an atom is an empty space.
2. The nucleus has a positive charge and nearly all the mass of an atom is concentrated in the nucleus.
3. The electrons orbit the nucleus at a distance and are much lighter than the nucleus.
4. The atomic structure is responsible for the properties of elements such as atomic mass and atomic number.
5. The atom is approximately 150,000 times bigger than the nucleus.
6. The discovery of protons and neutrons followed from Rutherford's model.

Limitations of Rutherford's model

1. The model didn't explain why electrons revolving around the nucleus do not lose energy and fall into the nucleus.
2. He did not completely address the nature of electrons and the way they occupied the vast space around the nucleus.
3. The model failed to explain the stability of atoms.

Qn:(a) The figure shows the path of three alpha particles moving towards a thin gold foil.

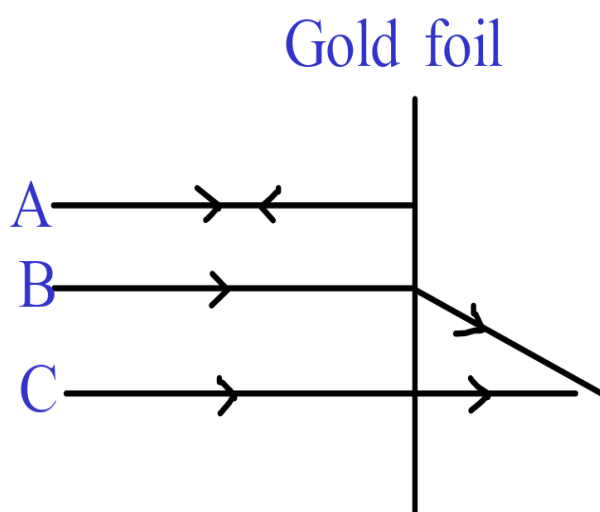


Particle A is moving directly towards a gold nucleus.

Particle B is moving along the line which passes close to the gold nucleus.

Particle C is moving along a line which does not pass close to a gold nucleus.

(b) Complete the path of the alpha particle A, B, C



Explain how the results of such as experiment using large number of alpha particles provides evidence for the existence of nuclei in gold atoms.

Particles which meet the nucleus are reflected through the same path and those which pass close to the nucleus are deflected through a large angle.

METHODS OF ELECTRON EJECTION

The process of removing an electron from an atom is called —textbfionisation.

Electrons can be ejected due to:

- (1) Thermionic emission
- (2) Photo electric emission

THERMIONIC EMISSION

Thermionic emission is the process by which electrons are ejected from a metal surface when heated.

Thermionic emission occurs in metals heated to a very high temperature. Metals have threshold temperature below which no electrons can be emitted called **threshold temperature**. Threshold temperature is the temperature below which an electron cannot be emitted from a metal surface.

QN How does the **surface area** of a metal affect the rate of thermionic emission?

The larger the surface area of a metal, the higher the rate of thermionic emission. This is because a larger surface area provides more space for electrons to be emitted.

QN Briefly discuss the effect of varying temperature on the rate of thermionic emission.

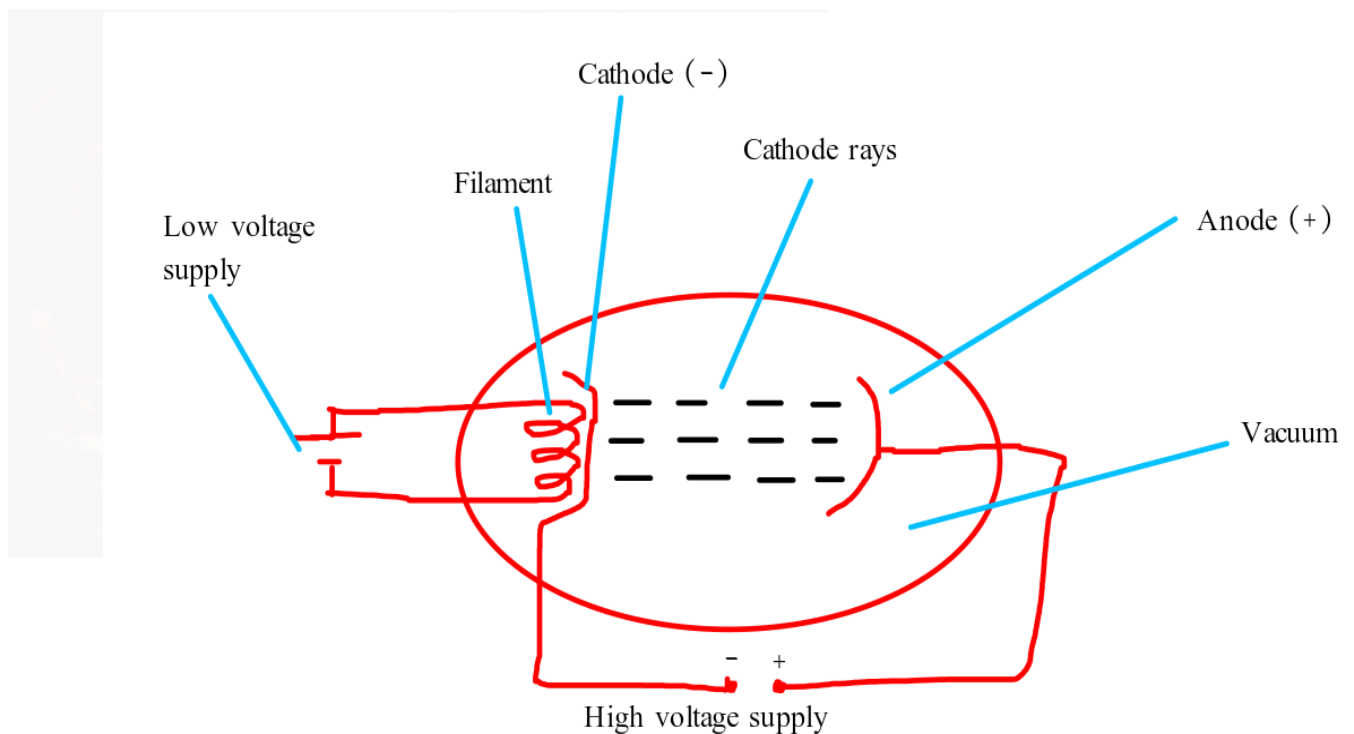
If the temperature is high, more electrons will be released because electrons gain more energy thus the rate of thermionic emission increases.

Thermionic emission is applied to in the following devices:

1. Thermionic diode valve
2. Cathode Ray Tube (CRT)
3. Cathode Ray Oscilloscope (CRO)
4. X-ray tube

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THERMIONIC DIODE VALVE



A **low voltage** supply heats the **filament** which in turn heats the **cathode** and electrons are produced **thermionically**.

When a **high voltage** is applied between the **cathode** and the **anode**, electrons are accelerated at a high speed from the cathode to the anode. This stream of electrons moving at a high speed is called **cathode rays**.

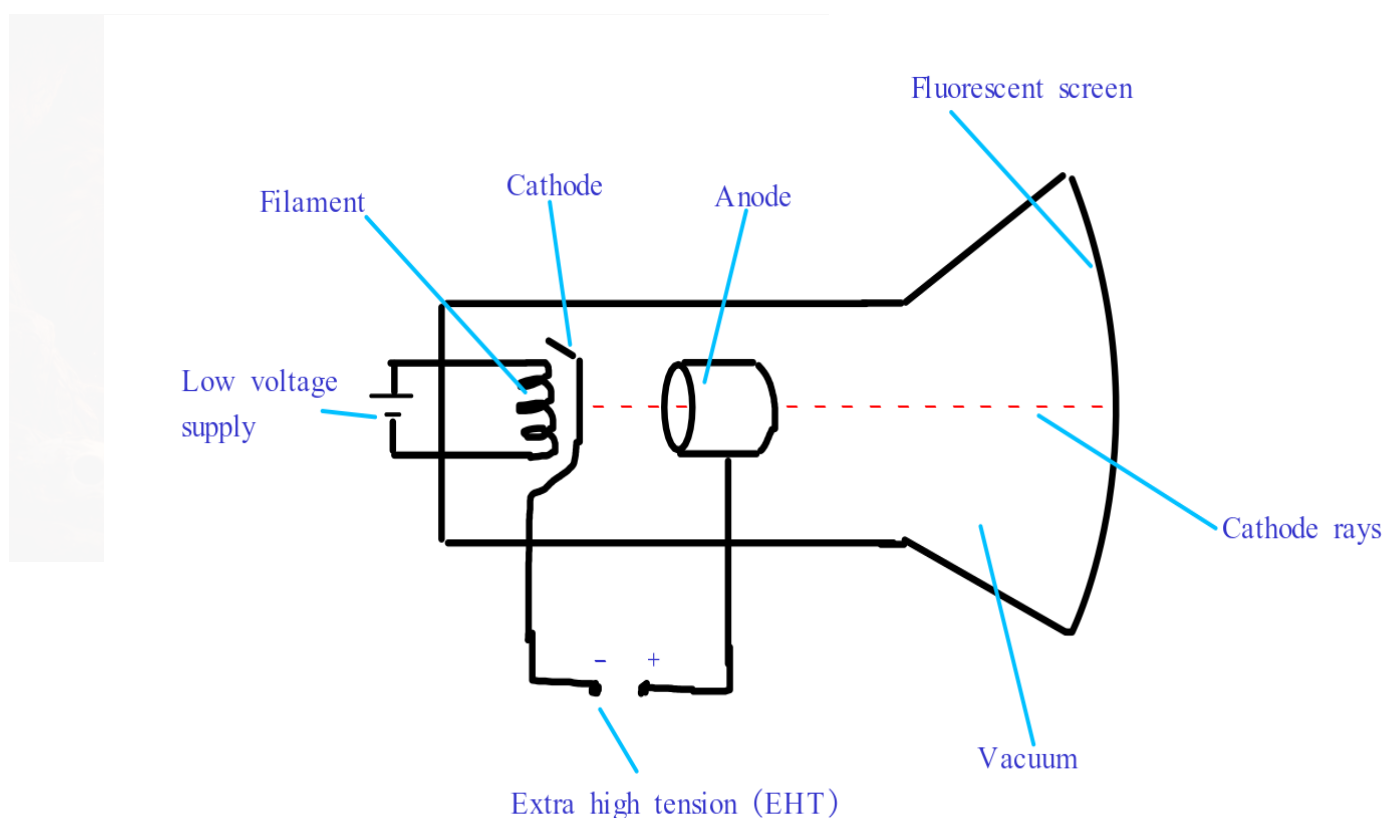
Definition: Cathode rays are streams of fast moving electrons.

NB: The air in the tube is evacuated (removed) so that there is no collisions between the electrons and the invisible air particles which would reduce the kinetic energy of the electrons.

If the accelerating p.d is increased the number of electrons ejected also increases.

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CATHODE RAY TUBE (CRT)



Low voltage heats the filament which in turn heats the cathode and electrons are emitted **thermionically**.

The high voltage applied between the cathode and the anode accelerates the electrons to a high speed towards the screen.

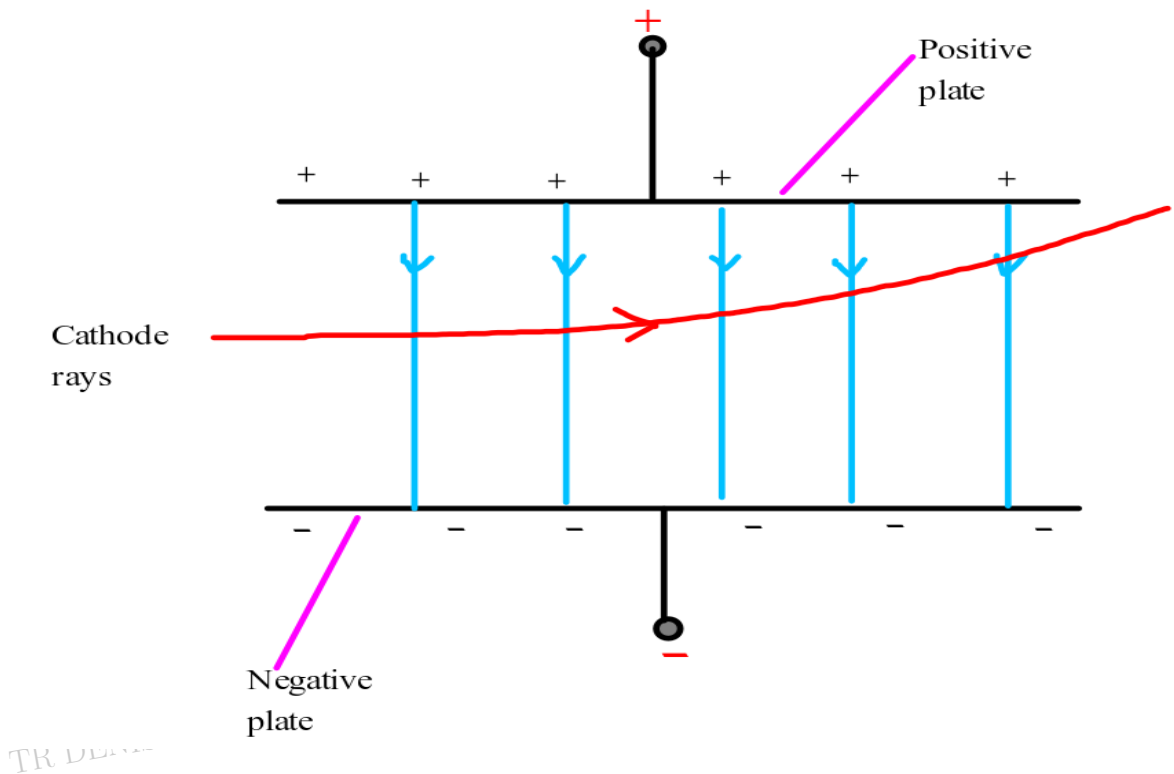
When the electrons strike the screen, they give up their kinetic energy and the screen glows (lights).

NB: The cathode ray tube is evacuated to prevent collision of electrons with invisible air molecules.

Properties of cathode rays

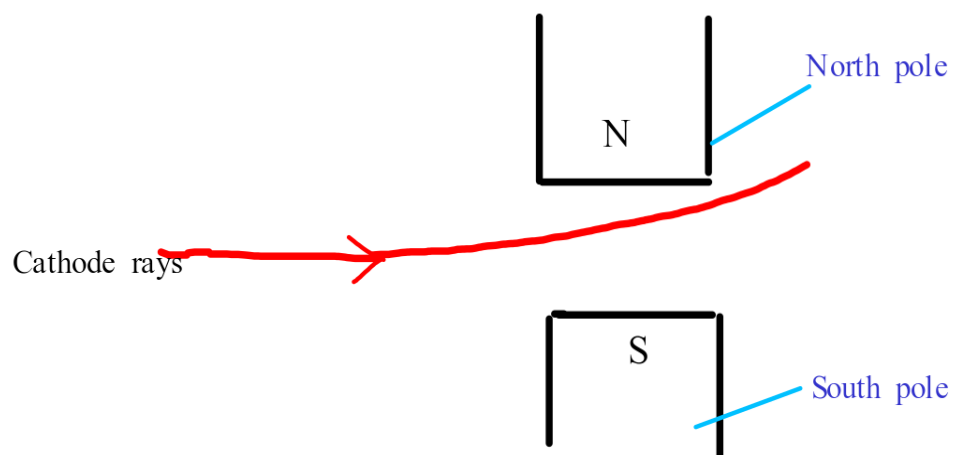
- (1) They are negatively charged.
- (2) They can be deflected by both electric and magnetic fields.
- (3) They produce X-rays when they strike matter.
- (4) They possess kinetic energy.
- (5) They ionise a gas through which they travel.
- (6) They cause fluorescence in certain substances.

Cathode rays in an electric field



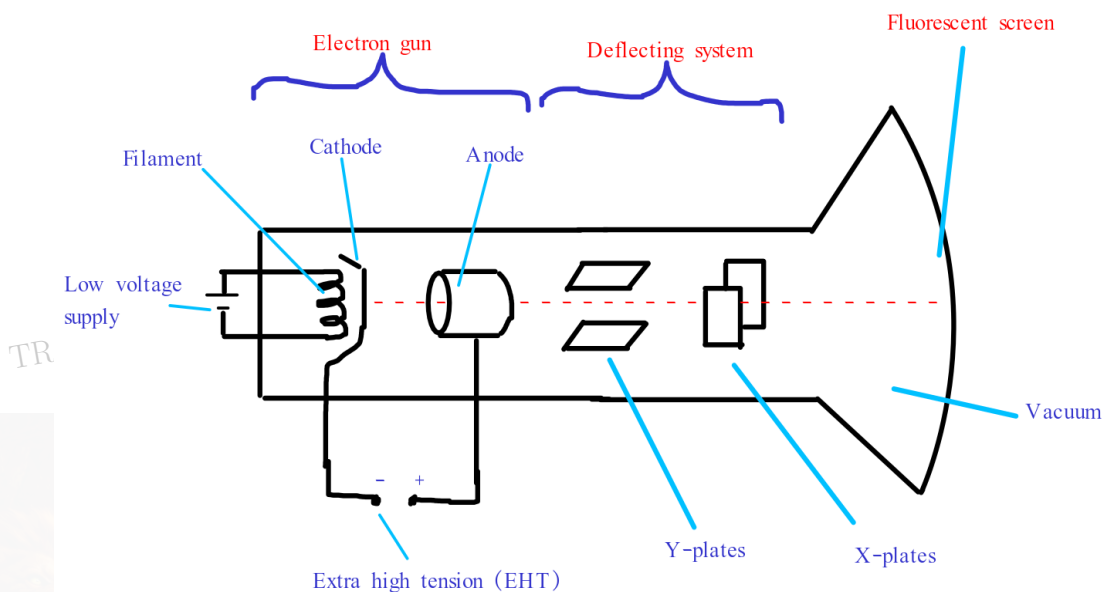
Cathode rays are directed midway between oppositely charged plates. The cathode rays are deflected towards the positively charged plates showing that they are negatively charged.

Cathode rays in a magnetic field



The cathode rays are deflected towards the **north pole** showing that they are negatively charged.

CATHODE RAY OSCILLOSCOPE (CRO)



A CRO is an instrument used to display/ study waveforms when a bright spot of electrons falls on the screen.

A CRO has three main parts: the electron gun, deflecting system and the fluorescent screen.

1. **The electron gun** This consists of a filament, cathode, control grid and the anodes.

The electron gun supplies electrons, accelerates the electrons and towards the screen and focuces the beam of electrons to the screen.

Functions of each part of the electron gun

Filament: It is used to heat the cathode.

Cathode: It emits electrons **thermionically** when heated.

Grid: It controls the number of electrons reaching the screen hence controls the brightness of the electron spot on the screen.

Anodes: THE anode accelerates the electrons produced and also focuses the electrons into a fine beam (spot) on the screen.

2. The deflecting system

This deflects the electron beam either vertically or horizontally. It consists of X-plates and Y-plates. When an input voltage is applied at the Y-plates, the spot deflects vertically across the screen.

3. The screen

When the accelerated electrons hit the screen, it glows into a bright spot.

Question: Explain the mode of operation of a CRO?

The low voltage heats the cathode and electrons are emitted **thermionically**. The electrons are accelerated and focused by the anodes towards the screen.

The X-plates deflect the electron beam horizontally.

The Y-plates deflect the electron beam vertically.

The screen glows to form a bright spot when struck by electrons.

NB

1. The glass tube is evacuated to prevent electrons from colliding with invisible air particles.

2. The graphite coating allows the earth.

Brightness control in a CRO

The brightness of a spot on a screen is controlled by **potential** applied at a grid.

When the grid is made more negative, few electrons crossover to the screen and the screen will be less bright.

When the grid is made less negative, more electrons crossover to the screen and the electron spot becomes more bright.

Uses of cathode ray Oscilloscope

Display waveforms.

Measures frequencies.

Measures small time intervals.

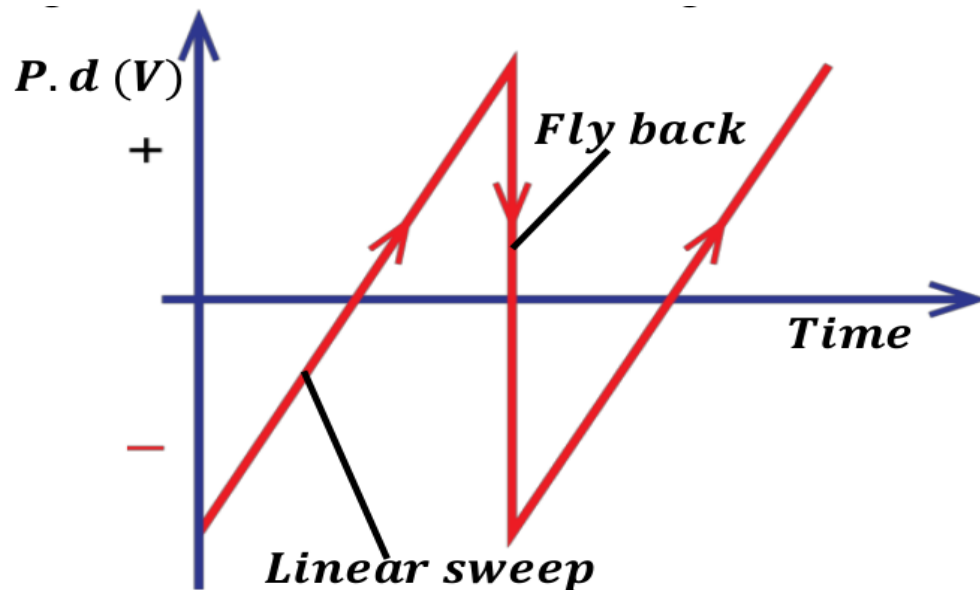
Measures wavelengths.

It measures both ac and dc voltage.

Question: What is a time base?

A time base is a special circuit that generates a pd which rises steadily to a certain value and falls steadily to zero.

The time base circuit is connected or applied on the X-plates. This generates a **saw tooth voltage** to the X-plate.



Waveforms produced in a CRO

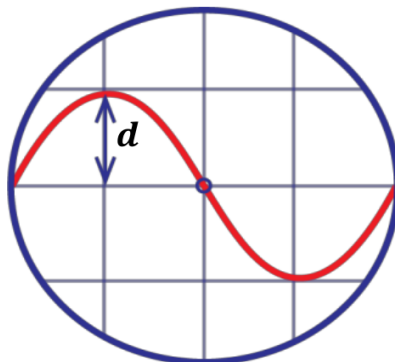
A CRO is used to study the voltage waveforms showing how pd varies with time. The waveforms appear different depending on whether the time base is on or off.

(i) No A.C voltage on the Y-plates and time base on 	(ii) A.C voltage on the Y-plates and time base off
(iii) A.C voltage on the Y-plates and time base on 	(iv) D.C voltage on the Y-plates only and time base off
(v) No D.C voltage on the Y-plates and time base on 	(vi) D.C voltage on the Y-plates and time base on
(vii) No potential difference is applied to the Y-plates and time base off 	

MEASURING VOLTAGE ON A C.R.O SCREEN

Voltage = [voltage gain/cm] x [amplitude]

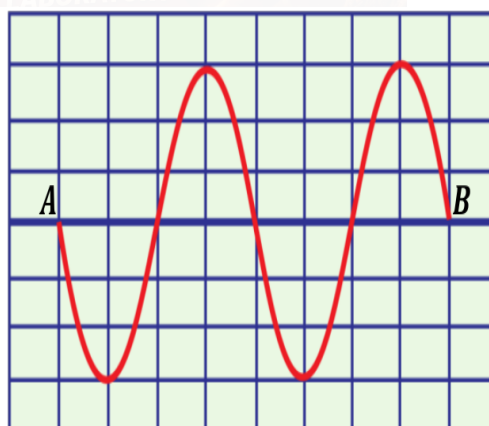
$$V = V_g d$$



Period T = [time base setting/frequency gain] x [wavelength]

Questions

1. A cathode oscilloscope CRO with time base switched on is connected across a power supply. The wave form shown in figure below is obtained and the length between each line is 1cm.



- a) Identify the type of voltage generated by the power supply.
- b) Find the maximum voltage (amplitude of voltage) generated if the

voltage gain is $5Vcm^{-1}$.

c) Calculate the frequency of the power source if the time base setting on the C.R.O is $5 \times 10^{-3}scm^{-1}$.

Solutions

(a) Alternating voltage.

(b)

From the graph, amplitude = 3 cm.

$V = \text{voltage gain} \times \text{amplitude}$

$$V = 5 \times 3$$

$$V = 15 \text{ V}$$

(c)

Period, $T = \text{Time base} \times \text{wavelength}$

wavelength = 4 cm

time base setting = $5 \times 10^{-3}scm^{-1}$

$$T = 5 \times 10^{-3} \times 4$$

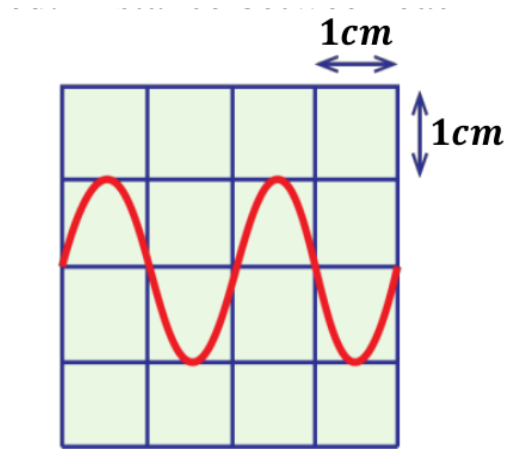
$$T = 0.02 \text{ s}$$

$$f = \frac{1}{T}$$

$$f = \frac{1}{0.02}$$

$$f = 50 \text{ Hz}$$

2. A CRO with the time base switched on is connected across a power supply. The wave form shown below is obtained. Distance between each line is 1cm.



- Identify the type of voltage generated by the power supply.
- Find the maximum value of the voltage generated if the voltage gain is 10 V cm^{-1} .
- Calculate the frequency of the power source if the time base setting on the CRO is $10.0 \times 10^{-3} \text{ s cm}^{-1}$.

Solutions

(a) Alternating voltage

(b)

Amplitude, $d = 1 \text{ cm}$

$$V = V_g \times d$$

$$V = 10 \text{ V cm}^{-1} \times 1 \text{ cm}$$

$$V = 10 \text{ V}$$

(c)

$T = \text{time base setting} \times \text{wavelength}$

$$T = 10.0 \times 10^{-3} \times 1.5$$

$$T = 0.105 \text{ s}$$

$$f = \frac{1}{T}$$

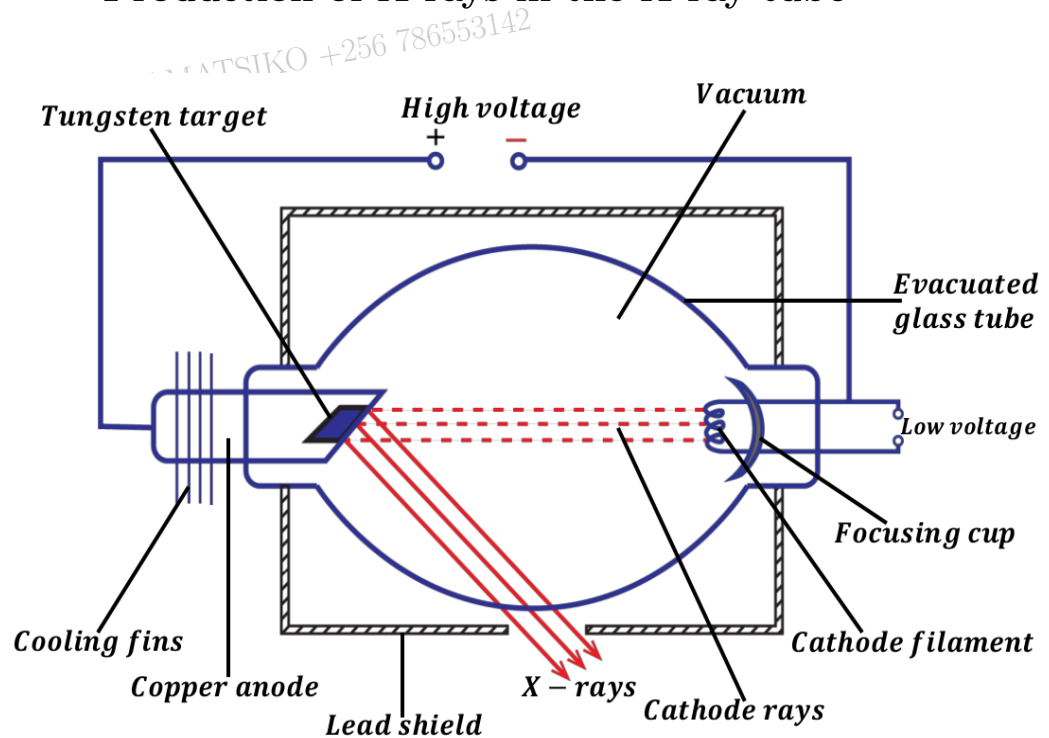
$$f = \frac{1}{0.015}$$

$$f = 66.7 \text{ Hz}$$

X-RAYS

These are electromagnetic waves of high frequency and short wavelength produced when fast moving electrons are stopped by a metal target.

Production of X-rays in the X-ray tube



The cathode is heated by a low voltage and electrons are emitted thermionically.

The emitted electrons are accelerated towards the metal target (anode) by the high pd between the anode and the cathode.

When electrons heat a metal target, they are stopped and X rays are

produced.

Only 0.5% of the kinetic energy of the electrons is converted to X rays and the rest is converted into heat.

NB: In the X ray tube,

1. The cathode is concave shaped to focus electrons on a metal target (tungsten).
2. The anode is made up of a good conductor of heat such as copper. This ensures that the heat generated when electrons hit the target is quickly conducted away from the target.
3. The target is made of tungsten because it has a high melting point therefore it withstands the heat generated without melting.
4. Cooling in the x ray tube is enhanced by the cooling fins outside the tube. The tube is highly evacuated to prevent electrons from colliding with air particles.
5. The X ray tube is surrounded by a lead shield to absorb stray X rays.

The energy changes that occur in the X ray in the production of X rays include:

Electrical energy $\rightarrow$ Heat energy $\rightarrow$ Kinetic energy $\rightarrow$ Heat + X rays.

Types of X rays

- (i) Hard X rays.
- (ii) Soft X rays.

(i) Hard X rays.

These are X rays that have a very short wavelength, high frequency and high penetrating power.

They are produced a **high** accelerating voltage that produces fast moving electrons.

(ii) Soft X rays.

These are X rays that have a relatively longer wavelength , lower frequency and low penetrating power.

They are produced by a **low** penetrating voltage which produces electrons moving at lower speed.

Question State the differences between hard X rays and soft X rays.

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Quality (hardness or softness) of X rays

The quality of X rays depends on the pd between the filament and the metal target.

The higher the potential difference, the harder the X rays produced.

Quantity / Intensity of X rays

The quantity of X rays produced depends on the filament current.

The higher the filament current, the more electrons emitted and the higher the intensity of the X rays.

Properties of X rays

They carry no charge

They are not deflected by both electric and magnetic field.

They travel in a straight line.

They travel at a speed of light ($3.0 \times 10^8 \text{ms}^{-1}$)

They cause fluorescence in some substances.

They penetrate matter.

They ionise a gas through which they travel.

Uses of X rays

Detect fractured bones and dislocations.

Locate foreign objects in the body.

Treat cancer/ Destroy cancer cells.

Sterilize medical equipments.

Detect lung tuberculosis.

Industrial uses

Detect cracks in metal casting and welded joints.

Used in study of crystals.

Inspect luggage for dangerous weapons at airports.

Dangers of X rays

Cause genetic mutations.

Cause cancer.

Damage eye sight.

They cause deep skin burns.

They destroy healthy living cells.

Safety precautions when dealing with X rays

Exposure to X rays should be limited to short time intervals.

X ray tubes should be surrounded by lead shield to protect operators from stray X rays.

X rays should be exposed to only body parts under investigation.

Question State the differences between Cathode rays and X rays.

Cathode rays	X rays
They are slower	They are faster
They are negatively charged	They have no charge
They are deflected by both electric and magnetic fields	They are not deflected at all
They are fast moving electrons	They are electromagnetic waves of short wavelength and high frequency

How x-rays are used to locate broken parts of a bone.

When X-rays are passed through the body onto the photographic plate or film, the bones which are composed of a much denser material than the flesh absorb most of the X-rays and appear white on the photographic plate or film.

The flesh which is composed of less dense material allows most of the X-rays to pass through it hence darkening the photographic film or plate. These shadows are studied in order to locate the broken part.

PHOTOELECTRIC EMISSION

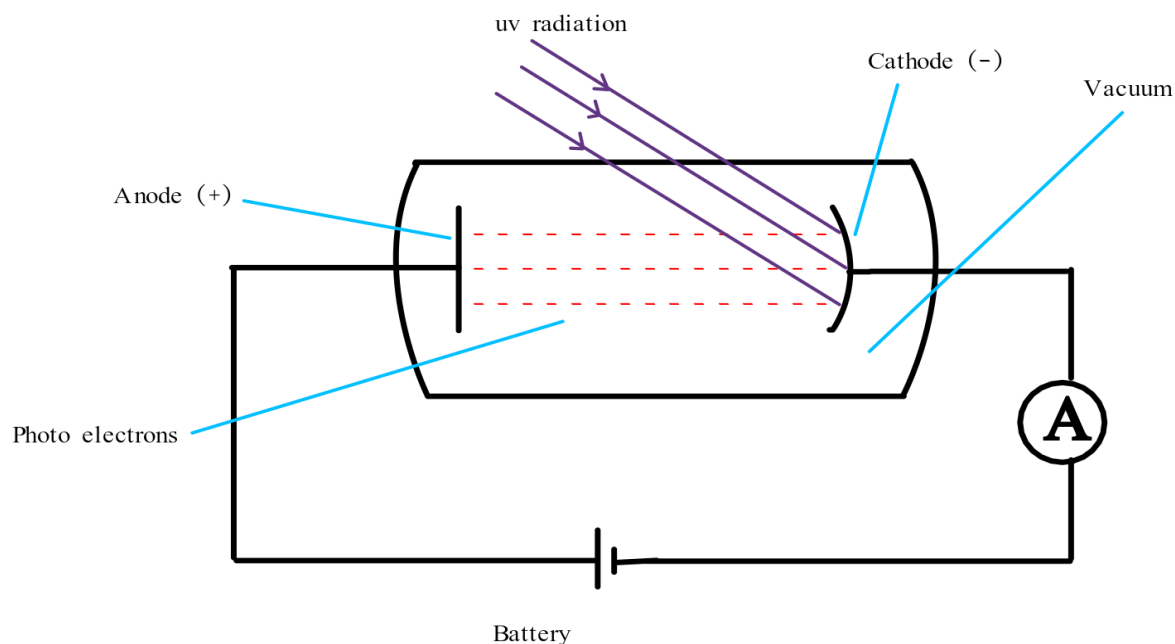
This is the process by which electrons are emitted from a metal surface when exposed to electromagnetic radiations of high frequency and short wavelength.

The electrons emitted are called photo electrons and the electromagnetic

radiations used are ultra violet (uv rays).

Photo electric emission usually occurs in photo electric cells (photo tubes).

Mechanism of photo electric emission



When uv rays fall on the the cathode, electrons are emitted by a process called photo electric emission.

The emitted electrons are then attracted to the anode, produce current in the curcuit causing the ammeter to deflect.

This shows that photo electric emission has occured.

NB: The tube is evacuated to prevent collision of electrons with air particles. The magnitude of current produced is directly proportional to the intensity of the radiations.

END